

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I, Y SRAVAN KUMAR (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SRAVAN Y

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview

1. Explain the difference between HDFS and traditional file systems.

HDFS is mainly used for storing very large amounts of data. It divides the data into blocks and stores them on different computers in a cluster. It also keeps copies of the data for safety. Traditional file systems usually store data on one computer or storage system.

In simple words: HDFS is for distributed storage, while traditional file systems are mainly for local storage.

2. Why is replication important in distributed storage systems?

Replication means keeping more than one copy of the same data. It is important because if one computer fails, we can still get the data from another computer.

For example, if there are 3 copies of a file and one server fails, the other copies are still available. So replication improves **data safety and availability**.

3. How does MapReduce divide work between map and reduce phases?

MapReduce divides a big task into smaller tasks.

In the **Map phase**, the input data is divided and processed by different mapper tasks. They produce intermediate key-value results.

Then, in the **Reduce phase**, the related results are grouped together and combined to produce the final output.

Flow:

Input → Map → Shuffle and Sort → Reduce → Output

4. What role does YARN play in large-scale Hadoop processing?

YARN is used to **manage the resources in a Hadoop cluster**. It decides how much CPU and memory should be given to different jobs.

It also schedules jobs and monitors them while they are running.

So, we can say **YARN manages the resources and jobs in Hadoop**.

5. Differentiate NAS and SAN in an enterprise data environment.

NAS and SAN are both used for storage, but they work differently.

NAS provides storage at the file level and is commonly used for sharing files over a network.

SAN provides block-level storage and is mainly used for applications that need high-performance storage, such as databases.

So, the simple difference is:

NAS → File-level storage

SAN → Block-level storage

6. What is the purpose of ETL in a big data pipeline?

ETL stands for **Extract, Transform, and Load**.

First, data is collected from different sources. Then it is cleaned and converted into the required format. Finally, it is loaded into a data warehouse or data lake.

For example:

Database + CSV files → ETL → Clean Data → Data Storage

ETL mainly helps to make the data **clean, organized, and ready for analysis**.

7. How does Apache NiFi help in data integration workflows?

Apache NiFi helps in **moving and managing data between different systems**.

It can collect data from databases, files, applications, and other sources. It can also transform, filter, and route the data.

One useful thing about NiFi is that we can visually create the data flow, so it is easier to understand and monitor.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Some common failures are **DataNode failure, NameNode failure, disk failure, and network failure**.

For DataNode failure, HDFS replication helps us access another copy of the data.

For NameNode failure, we can use **NameNode High Availability** with an active and standby NameNode.

We can also use monitoring and automatic recovery to detect failures and restart or move jobs when needed.

9. How would you design a secure data ingestion pipeline for a large organization?

I would first make sure that data is transferred through a secure connection. Then I would validate the data before storing it.

I would use **encryption, authentication, access control, and audit logs** to protect the data.

A simple architecture would be:

Data Sources → NiFi/Kafka → Secure Transfer → Data Lake → Access Control → Analytics

This helps protect the data while it is being transferred and stored.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

I would explain it with a simple example.

Suppose one person has a very large amount of work. Instead of asking one person to do everything, we divide the work among many people and let them work at the same time.

Hadoop works in a similar way. **HDFS stores data across different machines, YARN manages the resources, and MapReduce or Spark processes the data in parallel.**

If the amount of data increases, we can add more machines. That's why the system is called **scalable**.